

Continuous validation of product reliability in the series phase

Process description

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. Author, approver and reviewer are stored in and can be retrieved from the TERE system master data.

Amendments

The following amendments have been made to GS 90017:2011-08 und GS 90017 BBL: 2011-08:

- section 3 restructured;
- detailing of categories of reliability indicators;
- control loop for reliability indicators added;
- section 4.1 renamed and updated;
- sections 4.2 to 4.3 updated;
- section 4.4 added;
- sections 4.4 to 4.7 updated;
- section 6 deleted;
- section A incorporated into section 4.1 and updated;
- GS 90017 BBL integrated in Annex A;
- cited standards updated;
- Group Standard editorially revised.

Previous editions

GS 90017: 2005-11, 2011-08

GS 90017 BBL: 2005-12, 2011-08

1 Scope and purpose

This Group Standard defines a procedure to comply with the specified reliability in the series phase.

This Group Standard is applicable for all parts and components that are produced by order of BMW Group for use in series vehicles as well as their production processes.

The procedure defined in this Group Standard is not intended as replacement for the requalification acc. to GS 90018.

NOTE The responsibilities for tasks within the scope of this Group Standard are agreed in the respective Service Level Agreement (SLA). This Group Standard provides details of the contractual agreements with regard to the goods quality.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

DIN 40041	Dependability; Concepts
GS 90018	(all parts) Requalification of product and process at suppliers
GS 95024-3-1	Electric and electronic components in motor vehicles; Environmental requirements and tests
IATF 16949	Quality management system requirements for automotive production and relevant service parts organisations
VDA FMEA Handbook	Failure mode and effects analysis; Design-FMEA; Process-FMEA; Supplemental FMEA for monitoring & system response

3 Abbreviations and definitions

3.1 Abbreviations

BM	Special characteristics
DoE	Design of Experiment - statistic test planning
FEM	Finite Element Method
FMEA	Failure mode and effect analysis
GWK	Warranty and goodwill
HASS	Highly Accelerated Stress Screening
KQI	Customer quality index
LH	Requirements specification
MCS	Machine capability study
MnS	Months after start of production
MvS	Months before start of production
PFU	Process capability study
PMP	Problem management process
PQM	Product Quality Management
QI	Quality engineer
QMT	Quality management parts
QSH	Quality specialist in-house parts
SE	Simultaneous Engineering
SoP	Start of production
SPC	Statistical Process Control
Zuv	Reliability

3.2 Definitions

3.2.1 Reliability

Reliability is defined as the nature or characteristic of a unit with regard to its suitability to meet the reliability requirements during or according to defined periods at specified application conditions (DIN 40041).

Reliability is therefore a temporal extension of the term quality.

3.2.2 Reliability indicator

The task of the reliability indicator is to identify manufacturing-related alterations in the production process that may lead to a failure or defect of the component before the specified service life has been achieved.

Failures are

- an irreversible disturbance or malfunction,
- an occasional function disturbance or malfunction,
- impermissible wear.

For the reliability indicators, it shall be statistically proven that it is possible to derive a sufficiently precise statement on behavior over time. They shall be suitable to meet the reliability requirements in series production.

3.2.3 Categories of reliability indicators

Reliability indicators can be divided into the following categories:

Reliability-Stress-Test (acceleration test)

Test, e.g. stress and acceleration tests, endurance tests, environmental simulation. The stress shall be high but unrealistic defect patterns may not occur. The test shall be carried out until failure or until a defined degree of wear is measurable.

NOTE High expenditure, test duration usually too long for intervention, only random samples are possible.

Reliability quality characteristics

Non-destructive test: e.g. quality of a weld seam or soldered joint, size of shrinkage cavity (X-ray), number of defective locations.

Destructive test, e.g. tensile strength, fatigue strength, burst pressure, wear.

Reliability characteristics

These characteristics are defined directly in the drawing, e.g. part dimensions, surface finish, conductivity, internal resistance. They can be measured directly on the product.

Reliability process parameters

Process parameters, which have an influence on reliability. It is important that the process is understood and predictable in its context. This can be achieved for example by a DoE.

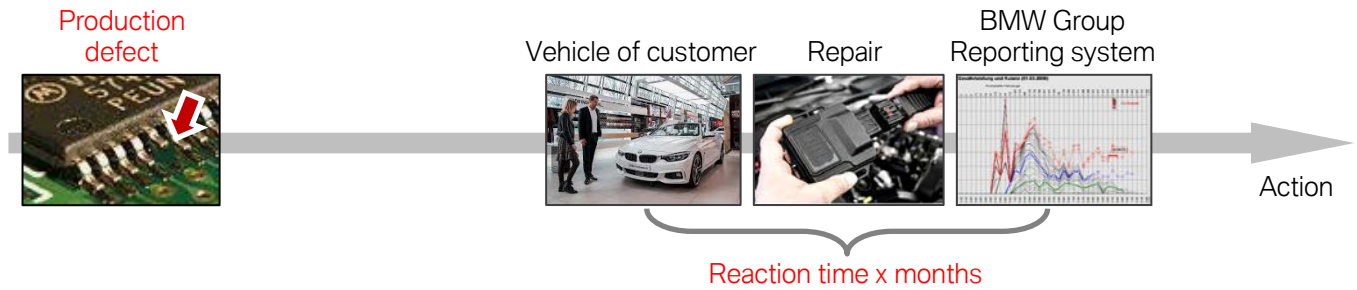
Table 1: Categories of reliability indicators

Category	Benefit	Expenditure	Reaction time	SPC-capable / 100 % Test	End of Line Test (HASS)	End of Line Test
Reliability stress test	very high	very high	poor	no	-	random sample
Reliability quality characteristics	very high	high	medium	only non-destructive	only non-destructive	-
Reliability characteristics	high	low	fast	possible	possible	-
Reliability process parameters	high	low	very fast	possible	-	-

3.2.4 Fast corrective control loop

The following is a simplified comparison of the control loop with and without reliability indicators.

Control loop without reliability indicator



Fast corrective control loop with reliability indicator

Figure 1: Control loop without reliability indicator

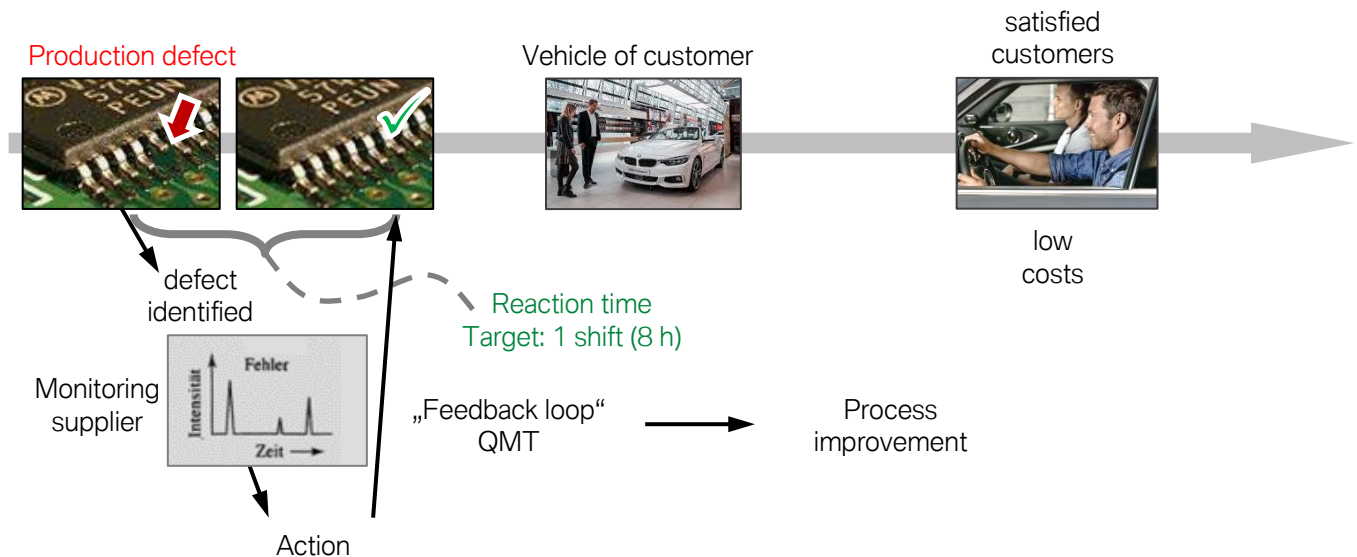


Figure 2: Fast corrective control loop with reliability indicator

4 Process

4.1 Procedure

Simplified process description of reliability indicators:

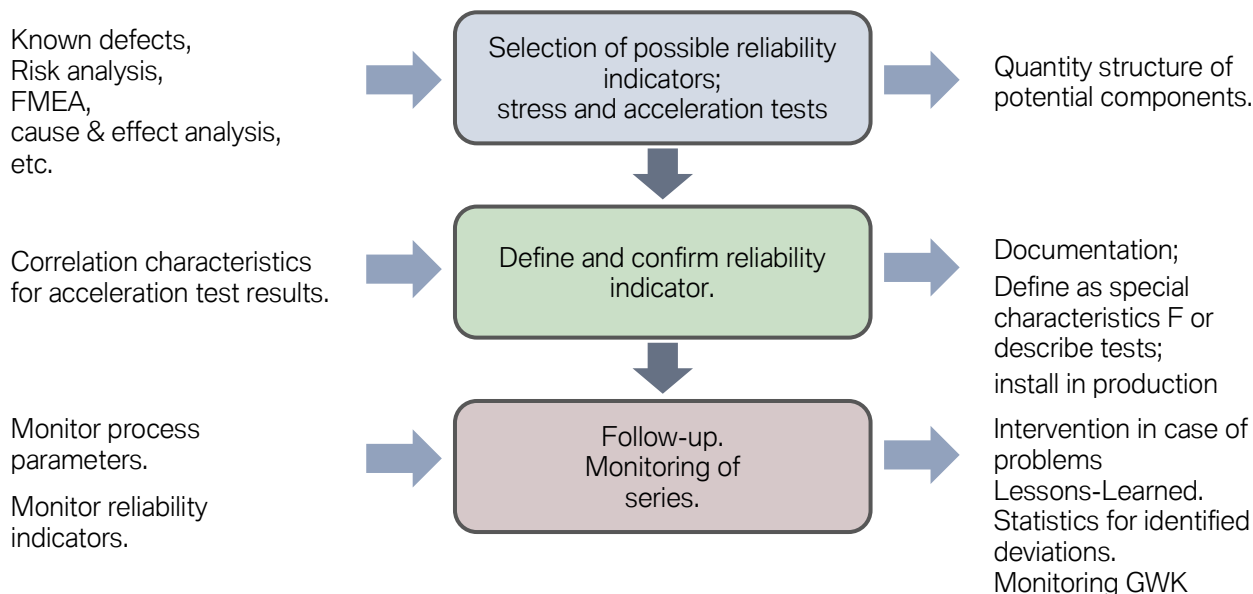
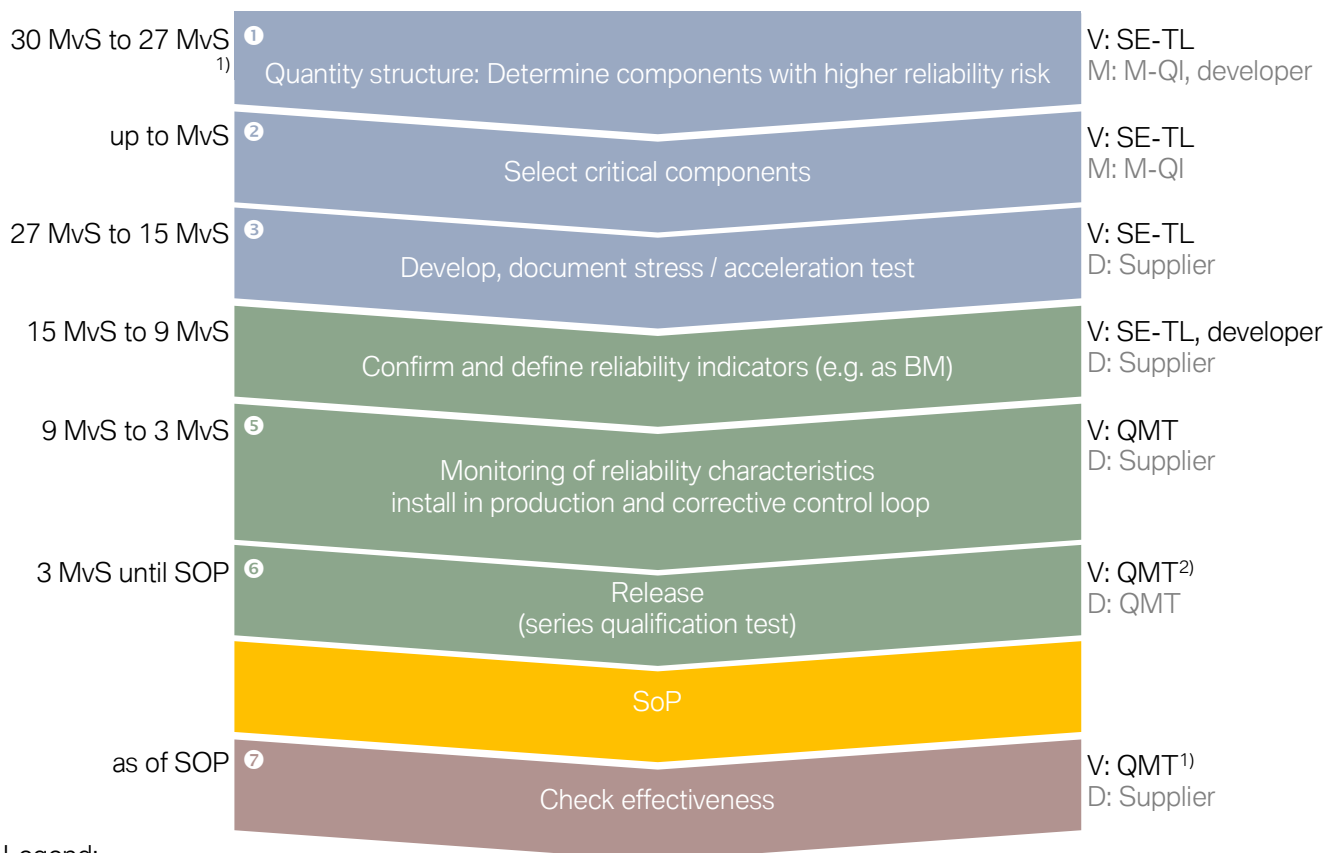


Figure 3 Process of reliability indicators

The responsibilities for the subsequently described activities are shown in Figure 4. The responsible persons shall be involved in the process from the time of first nomination.



Legend:

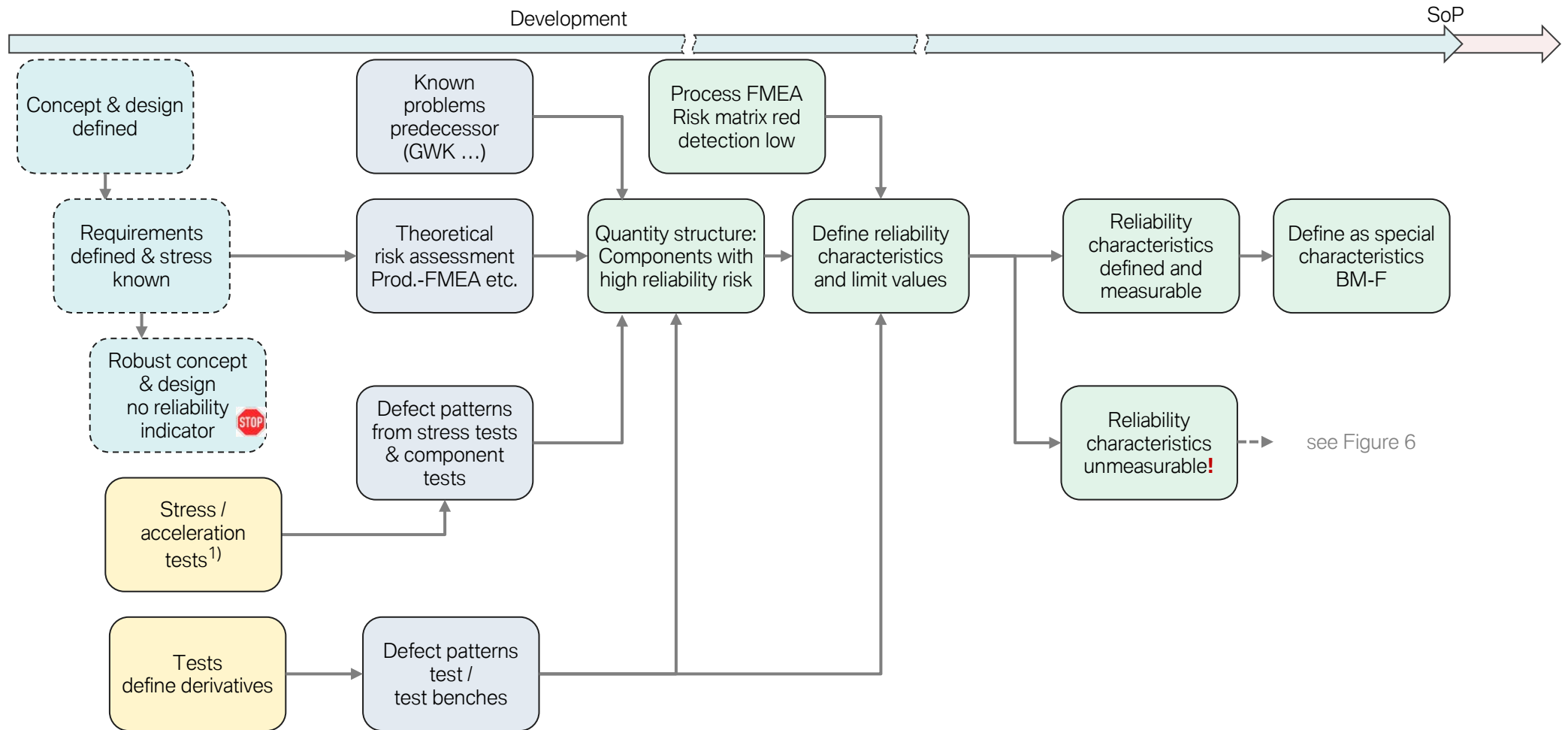
V responsibility
D implementation
M collaboration

1) for in-house parts → QSH

2) QSH

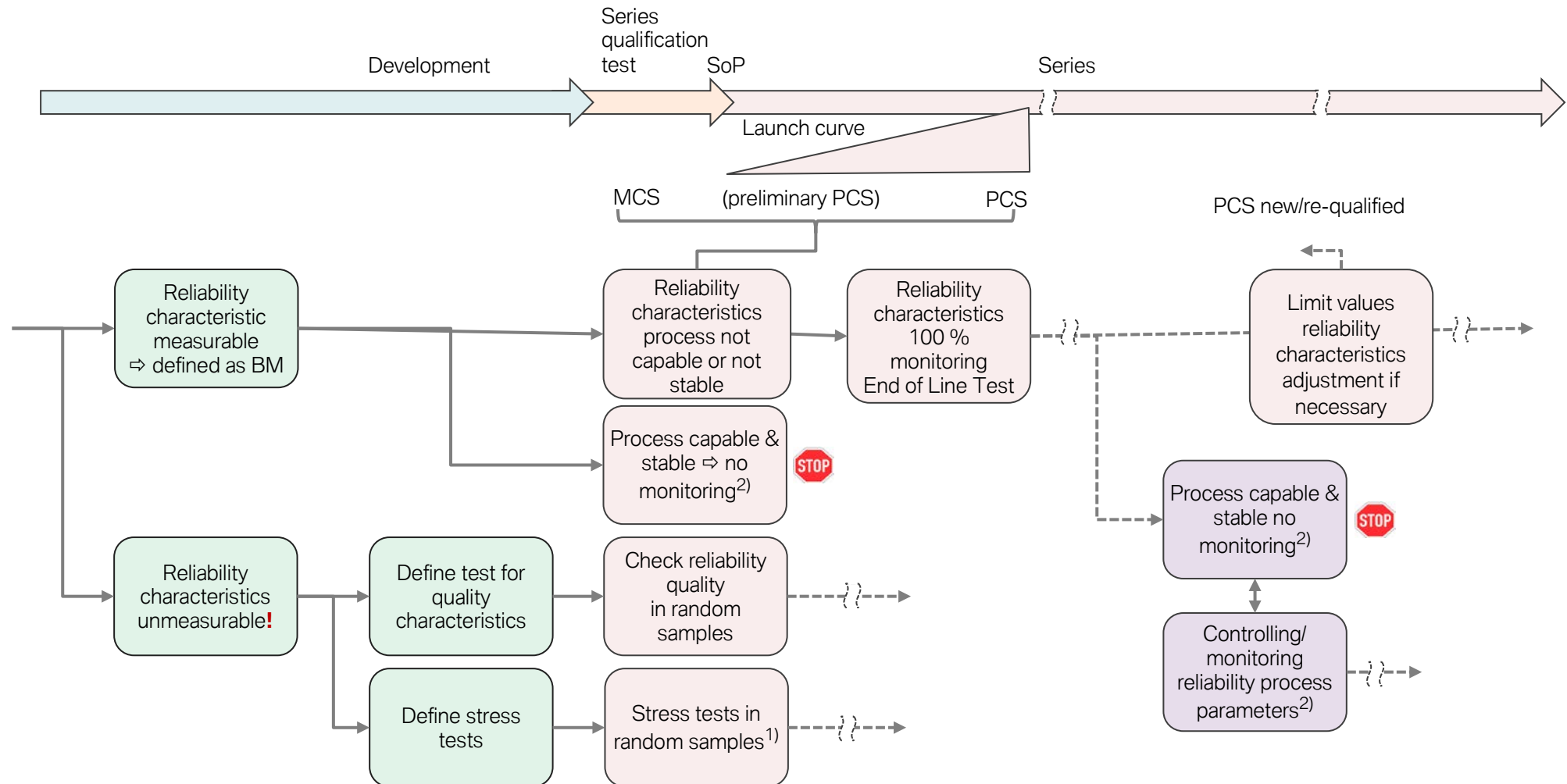
Figure 4: Responsibilities

The process for reliability indicators is described as follows:



1) e.g. from GS 95024-3-1 or other specifications & standard tests

Figure 5: Process of reliability indicators for ItO



1) Larger cycles e.g. after process changes and requalification.

2) Reliability characteristic is only checked if the process has changed. Monitoring of process parameters is also an option.

Figure 6: Process of reliability indicators for OtD

By request of BMW Group the supplier shall provide proof of

- planning of continuous reliability validation (derivation of reliability indicators and definition of suitable validation tests) and
- their implementation in production (installation of validation test in the production process and establishment of a fast corrective control loop).

For the planning and steering of the implementation process the evaluation sheet in Annex A should be used by the responsible development department.

For involvement of the supplier in the entire process the responsible development shall carry out a kick-off event in good time (see evaluation sheet in Annex A).

4.2 Risk assessment

The first step at the start of the project is to determine the reliability risk of the component by the engine and/or vehicle project. In this context mainly reliability risks shall be taken into account if an incapable or instable production process is to be expected.

For each component a risk assessment shall be performed. The following criteria shall be assessed at least:

- effect of an early failure of the component on the total vehicle (significance and severity of defect);
- if one parts already has a high failure rate in the (warranty graphs and warranty issues);
- if the production process is not capable or stable. This is to be expected if a part is produced in a new production location or with a new production process;
- if a new supplier is nominated;
- if a new and/or innovative technology is used (both product and process);
- if a part is highly complex;
- if adequate testing and validation is not possible.

4.3 Identification of the reliability indicators

The identification of reliability indicators is possible via actual defects or on the basis of theoretic risk considerations.

On the basis of actual defects:

- warranty analysis (predecessor, component with comparable concepts);
- findings from tests, endurance tests, customer-oriented test, etc.;
- findings from stress and acceleration tests;
- lessons learned from predecessor projects;
- known PQM points, 8D reports, problem-solving sheets;
- experience and expertise of the supplier.

On the basis of theoretical risk assessment:

- risk analysis (risk filter, risk matrix, FMEA);
- identification of relevant topics from P diagram (especially control factors and part to part);
- cause and effect analysis for selected risk topics:
 - fault tree;
 - by calculation e.g. FEM, vibration simulations.

In concrete cases, Weibull analysis of field failures of comparable components are possible. In a theoretic approach, system analyses or FMEA can be used in case of new systems. This information facilitates the determination of failure mechanisms (cause and effect correlations) and the deduction of potential influencing factors of a failure and the ensuing suitable reliability indicators.

In the following the risk assessment by means of FMEA is to be described more closely.

The product FMEA refers to product properties and characteristics and their possible malfunctions. The process FMEA identifies possible process relevant influences. Due to the multitude of topics in an FMEA it is necessary to concentrate on the key topics. The risk matrix is useful in this case (see FMEA handbook). The potential reliability indicators are selected primarily via their probability of occurrence A during operation.

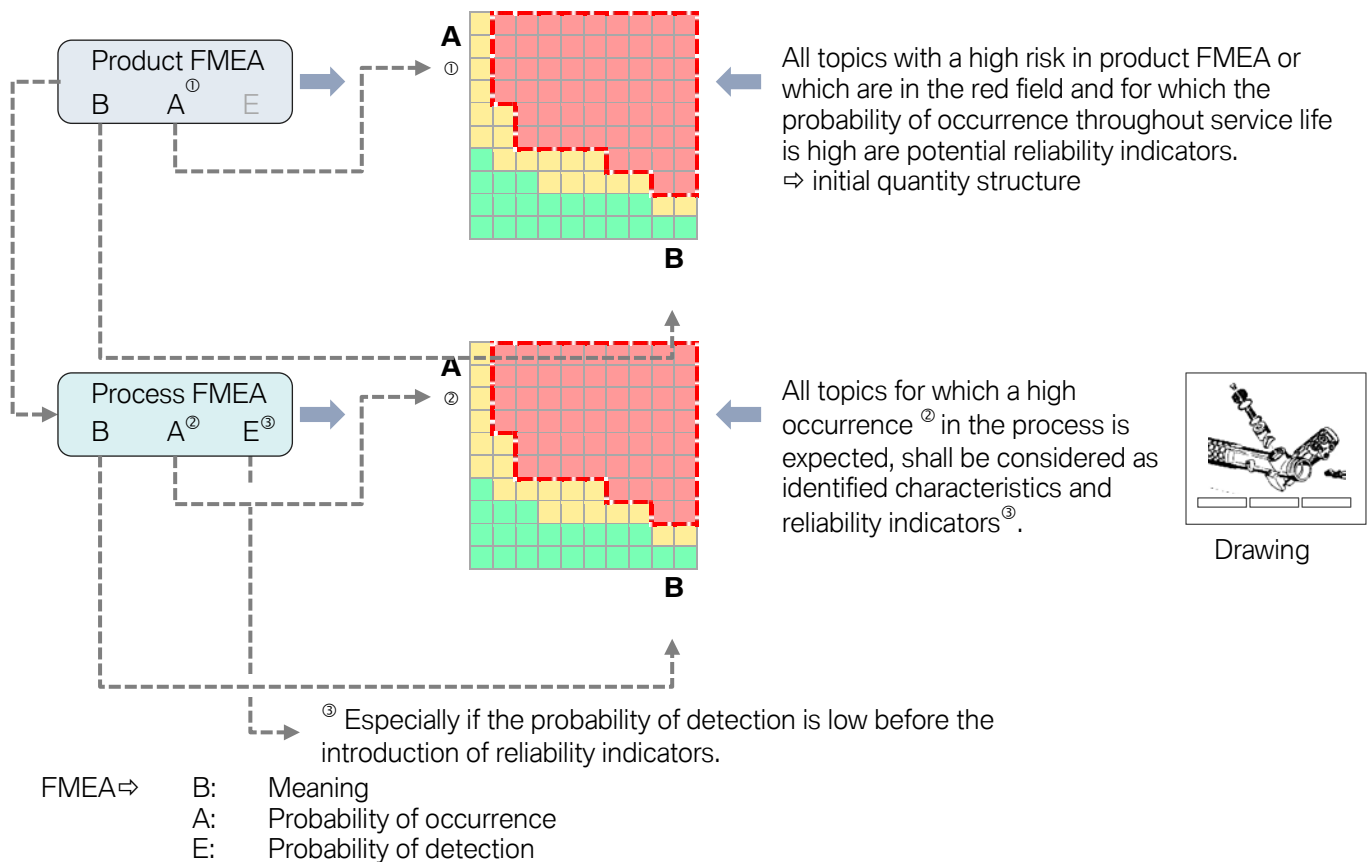


Figure 7: Risk matrix from FMEA

NOTE Value 9 and 10 are safety-relevant and therefore already special characteristics that are being monitored. The priority for processing is determined by the expected probability of occurrence A. In case of the process, probability of occurrence in the process is the determining factor.

How are reliability indicators determined with FMEA?

All significant topics in the yellow and red fields shall be tested if probability of occurrence refers to the operation (e.g. service life, aging).

If these potential reliability indicators are also critical in the process FMEA, especially if their detection is low, these reliability indicators shall be used and monitored.

How are they marked in the FMEA?

The marking as reliability indicator via special characteristics can be defined directly as F-marking (form classification column K). In this way they will be monitored by a proven process with defined responsibilities.

In the remark field of an FMEA element, additional information can be entered, e.g. links to documents or information about why this is a reliability indicator.

The workflow for definition of reliability indicators and the follow-up are filed in the system in the KQI.

4.4 Selection of the suitable validation of the reliability indicator

As regards the indicators intended for reliability assurance suitable test processes shall be defined which allow for a statement regarding fulfillment of function over time.

The scope and frequency of tests shall be designed in such a way that an impermissible deviation of a monitored indicator is detected so that it is possible to react quickly to changes in the productions process.

The reaction to unacceptable deviations in the manufacturing process and the time slot available for the reaction are a function of the severity of the impacts of a premature failure in the vehicle. For the planned test procedure, the following can be deduced,

- which quality of fault detection is required (test evidence) as well as
- period after which test results shall be available.

For all tests action limits shall be defined. If these limits are exceeded or fallen short of the fast correction control loop at the supplier shall be triggered.

In case of changes on the design or the process the indicators and their action limits shall be checked and adapted, if required.

4.5 Identification of the reliability indicators

Identified reliability indicators with model characteristic shall be defined as special characteristics F, see section 4.3 section FMEA.

In addition to special characteristics it is also possible to mark tests on the drawing. This concerns model reliability stress test and reliability quality characteristics.

For example, this can be done by means of a rhombus see Figure 8.

Reliability indicators marked in this way shall be summarized in a table with clear and countable numbering. Figure 9.

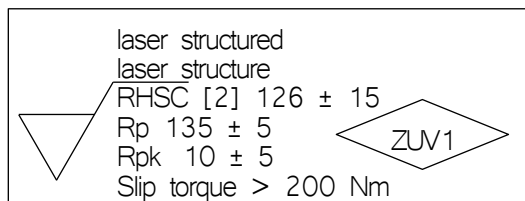


Figure 8: Marking of characteristic on the drawing (example)

Marking	Merkmal	Characteristic	Coordinate
ZUV1	Durchrutschmoment Laserstruktur	Slip torque laser structure	B14
ZUV2	Oelleckage	Leak test with oil	F1 – F2
ZUV3	Hysterese in Oel	Hysteresis in oil	E3 – F3
Quantity: 3			
Reliability characteristics according to: BMW GS 90017			

Figure 9: Table with description of tests (exemplary)

4.6 Preparation of the fast corrective control loop

The fast correction control loop, see Figure 2, shall be developed by the supplier with support of the QMT specialist or the quality specialist for in-house parts until start of production (SoP) and shall be integrated into the manufacturing preparation.

Depending on the identified reliability risk it shall be defined how manufacturing batches that exhibit unacceptable deviations are treated at the supplier, and how the responsibilities are distributed.

4.7 Validation and release

Until pre-series production proof shall be furnished that the planned validation test reliably detects production instabilities with impact on reliability using the specified number of random samples.

It is recommended to assign a sufficient number of “poor parts” into the test to simulate an impermissible change in the production process.

The characteristics of a poor part can be described by means of the completed analyses for identification of the reliability indicator.

If the assigned poor parts are not detected, a systematic cause analysis with traceable documentation shall be carried out so that the test can be subsequently revised accordingly.

The release of reliability indicators and test procedures for reliability validations shall be carried out by the responsible development department. The test shall be validated by the responsible employee of the BMW Group QMT or the QSH. Documentation of the test validation is mandatory and shall be approved by QMT or QSH.

The validation shall be checked at suitable intervals after SoP.

4.8 Handling of defects in series production

Generally, the supplier is responsible for immediate problem solving within the fast corrective control loop. If problems are identified, they shall be reported to BMW Group immediately.

If it is necessary to transfer problem solving to BMW this shall be done in compliance with the rules of the BMW problem management process (e.g. supplier cannot solve the problem without support, measure not introduced, schedule not adhered to).

If feedback is provided via the control loops installed by BMW with regard to production-related reliability problems in the field, the supplier shall carry out a systematic cause analysis with traceable documentation (control loop).

4.9 Effectiveness test

At regular intervals, the responsible QMT or QSH employee shall check by means of feedback about the situation in the field whether it is necessary to adjust the defined reliability indicators.

The time of the test is to be defined by the QMT employee or the QSH following processes described in GS 90018 (all parts).

5 Documentation.

The results of above process shall be documented comprehensively and coherently by the supplier. The form of documentation is to be agreed with the responsible BMW development department.

At least the following contents shall be documented:

- possible failures and their causes (potential reliability indicators);
- description of corresponding failure mechanisms (correlations);
- reliability indicators specified for validation;
- test incl. test specification;
- intervention limits;
- used measuring systems, resolution, proof of gauge capability (e.g. R&R value).

The test specification shall comprise at least the following:

- test load (e.g. temperature level, temperature gradient, number of cycles, ambient pressure);
- limit values;
- test frequency;
- test duration;
- random sample scope;
- reliability of result or fault probability;
- confidence range.

A (Annex normative) Assessment sheet



**Assessment sheet for monitoring of
Continuous validation of product reliability
in the series phase**

Supplier (Company, production site)	
Project (Vehicle, engine, component)	
Contact at supplier (Name, Department)	
SE team leader BMW Group (Name, Department)	
QI BMW Group (Name, Department)	
QMT BMW Group (Name, Department)	
Q-Specialist in-house parts (Name, Department)	
Evaluation by (Name, department, signature)	
Date of evaluation	

The questionnaire comprises the following topics:

1. Organization / boundary conditions / procedures
2. Reliability indicators
3. Validation tests
4. Implementation in series production

Each topic contains separate questions that shall be answered as follows:

- +** Criterion fulfilled
- O** Criterion partly fulfilled
- Criterion not fulfilled

Criteria that are not completely fulfilled shall be furnished with remedial measures and corresponding scheduling.

Time	Action	Deadline, planned	Comment / Description	Date, completed
30 to MvS	Determine components with higher reliability risk			
up to MvS	Select critical components			
up to 15 MvS	Develop, document stress / acceleration test			
up to 9 MvS	Identify possible reliability indicators and define as special characteristics			
up to 9 MvS	Develop and document validation test			
3 MvS	Install and validate corrective control loop			
3 to 0 MvS - SoP	Release series qualification test			
as SoP, for the first time approx. 4 MnS	Check effectiveness revalidation			

1. Organization / boundary conditions / procedures		+	O	-	Comment / Remark
1.1	Was a kick-off meeting held in the SE-Team?				
1.2	Is the objective of continuous validation of product reliability in the series phase known and are all persons involved suitably qualified and have they been trained for implementation?				
1.3	Is there a schedule for further procedure? Are all milestones of the process known? Is this schedule available and used as the planning basis for monitoring?				
1.4	Are all tasks, competences and responsibilities defined clearly and unambiguously and agreed upon? Has a central contact (single point of entry) been nominated for BMW?				
1.5	Is the design consolidated to an extent that planning of Continuous Assurance of Product Reliability can be initiated? Have all design decisions been made?				
1.6	Are all requirements related to reliability defined clearly, documented and known (e.g. in the requirements specification)? What function, respectively characteristic of the function can be fulfilled reliably over what time and under what operating conditions? What is the minimum reliability that has to be proven with which failure probability?				
1.7	Has the process of continuous validation of product reliability in the series phase been transferred from the supplier to the main sub-suppliers?				
1.8	Has it been determined how the derivation of the reliability indicators and the associated validation tests are to be documented (e.g. independent documentation list or familiarization with process FMEA)?				
1.9	Does the status of documentation correspond to the current status of product and process development?				

2. Reliability indicators		+	O	-	Comment / Remark
2.1	Have all field data and risk analyses been evaluated?				
2.2	Have all failure pattern been identified (destruction of component, irreversible function deviation or wear beyond the permitted extent)?				
2.3	Have all reliability indicators been determined with the help of a systematic procedure on the basis of the failure patterns? Have all relevant reliability indicators been determined?				
2.4	Are all related failure mechanisms known and traceably documented? (Depiction of chain of effects with at least 3 steps, e.g. inadequate thermal connection → poor heat discharge → overheating → failure of electronics)				
2.5	Can the failure causes from the past be applied to the current component?				
2.6	Is there any proof that the potential reliability indicators can be used to derive a sufficiently accurate statement with regard to compliance with reliability requirements (correlation with reliability)?				
2.7	Is there an evaluation scheme for evaluation of failure patterns and reliability indicators in relation to each other? Is it assured that the most representative reliability indicators are selected for validation?				
2.8	Are the reliability indicators documented in the release records applicable for the respective component?				

3. Validation tests		+	0	-	Comment / Remark
3.1	Is the planned test process suited to assure the reliability indicator (e.g. at accelerated endurance run: no effect of acceleration on failure mechanism)? Is the test procedure suitable to make a statement on the fulfillment of functions over the time (service life)?				
3.2	Has it been proven that a deviation from the target condition can be reliably detected by the test? (Proof that, with a given scope of specimens, deliberately introduced "reject parts" are detected reliably).				
3.3	Should continuous measuring values be recorded for determination of scatter and error share of the process (depiction of trends, no good/poor statements)?				
3.4	Has a complete test specification been prepared? (Contents: test load, limit values, test frequency, test duration, scope of random samples, confidence level / error probability / confidence interval)				
3.5	Is the test specification a controlled document (document number, release date, version)?				

4. Implementation in series production		+	O	-	Comment / Remark
4.1	Has the test been validated, approved and released for series monitoring (by the responsible QMT employee or Q specialist for in-house parts)?				
4.2	Does the supplier have clear test instructions (work instructions with description of test procedure; in the form of a separate document or a part of the test specification)?				
4.3	Is documentation and statistic evaluation of test results planned?				
4.4	Has an agreement been reached between BMW Group and the supplier on how the test results and their evaluation are to be passed on the BMW Group (e.g. reporting line, reporting interval, data format)?				
4.5	Is there a clear definition of how to deal with production batches if the reliability indicator violates the defined limit values? (Are the control loops clearly described? Is there a production control plant according to IATF 16949, that contains the defined reliability indicators? Is the connection to BMW-PMP assured?)				
4.6	Have all planned tests and their documentation been started and is the reporting line active? Are they are an integral part of series-accompanying quality tests?				
4.7	Is it assured that reliability indicators and validation tests are adjusted in case of product and process changes?				